EFU FOOD PROCESSING LICENCE v1.0

Experimental EFU-based Food Processing & Sovereignty Framework

0. Purpose and Scope

Experimental EFU-based licensing and measurement framework for food processing industries (meat, dairy, bakery, canning, oil processing).
NOT a legal requirement, HACCP substitute, or food safety certification replacement, but an open research protocol measuring food processing resource use in human-scale EFU metrics (water, energy, raw material waste, emissions).

I. Core Concepts

1. **EFU-Food (EFU-FOOD)**
   EFU framework application for food processing industries:
   - **EFU-H_FOOD**: energy (baking, cooking, refrigeration, drying, pasteurization)
   - **EFU-W_FOOD**: water (washing, blanching, steam, cooling circuits)
   - **EFU-M_FOOD**: raw material loss (30-50% food waste)
   - **EFU-C_FOOD**: carbon (methane from dairy, N2O from fertilizers)
   - **EFU-S_FOOD**: entropy/heat (refrigeration cycles, sterilization)
2. **Food Processing Sovereignty Gap (SS_FOOD)**
   SS_FOOD = TEFU_FOOD_total – Regional Food Carrying Capacity (RFCC_FOOD)
   Measures if processing stays within regional food security regeneration capacity.
3. **MROI_FOOD**
   Food processing Metabolic Return on Investment = THI_FOOD / TEFU_FOOD
   THI_FOOD = calorie coverage + local food security + jobs + nutrient density.

II. EFU Food Processing Licence – Research Status

The licence documents EFU-audited food processing plants with standardized metrics.
Does NOT replace HACCP, ISO 22000, or food safety authority permits.
Enables bakeries, dairies, meat processors benchmarking in EFU research context.

III. Metrics and Thresholds (Food processing-specific)

1. EFU Components (per kg finished product)

EFU-H_FOOD (Energy)
EFU-H_FOOD = processing_energy_MWh / 12.88 / finished_product_kg
Example: Bread = 0.8 kWh/kg → 0.8/12.88/1 = 0.000062 human-years/kg

EFU-W_FOOD (Water)
EFU-W_FOOD = water_m3 / 18.25 / finished_product_kg
Example: Milk = 5 liters water/liter → 5/18.25/1 = 0.274 human-years/kg

EFU-M_FOOD (Raw material loss)
EFU-M_FOOD = waste_kg / 30.0 / finished_product_kg
Example: 40% loss → 0.4/30/1 = 0.00133 human-years/kg

TEFU_FOOD = 1.0×EFU-H + 1.5×EFU-W + 1.2×EFU-M + 0.5×EFU-C + 0.8×EFU-S

2. Regional Food Carrying Capacity (RFCC_FOOD)

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RFCC_FOOD = (Sustainable_calories_MJ_year / Population) / 10,500 MJ/person/year baseline
Example: Netherlands = 2,500T kcal / 17.8M / 10,500 MJ = 0.134 EFU_FOOD/person/year
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IV. Licence Categories (EFU-FOOD Labels)

Level	SS_FOOD Condition	MROI_FOOD	Label	Interpretation
1	SS_FOOD ≤ 0	≥1e-3	**EFU-FOOD SOVEREIGN**	Within food security capacity
2	0 < SS_FOOD ≤ 5k EFU	≥1e-4	**EFU-FOOD BALANCED**	Monitorable
3	5k < SS_FOOD ≤ 25k EFU	≥1e-5	**EFU-FOOD DEFICIT**	Overloaded
4	SS_FOOD > 25k EFU	<1e-5	**EFU-FOOD CRITICAL**	Structurally imbalanced

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V. Regional RFCC_FOOD Calibration Examples

Country/Region	Calories (T kcal/year)	Population (M)	RFCC_FOOD (EFU/person/year)
NETHERLANDS	2,500	17.8	0.134
USA	150,000	340	0.042
INDIA	120,000	1,430	0.008
CHINA	500,000	1,410	0.034
EU AVERAGE	1,200,000	448	0.025

VI. Example: Dutch Dairy Processing Plant

Input (annual plant):

- Production: 50M liters milk (50,000 tonnes)
- Water: 250M liters (250,000 m^3)
- Energy: 25,000 MWh
- Staff: 300 workers
- Waste: 15,000 tonnes dairy products
- Region: Netherlands (RFCC_FOOD = 0.134 EFU/person/year)

Calculation:

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EFU-W_FOOD = 250,000/18.25/50,000,000 = 0.0274 human-years/kg
EFU-H_FOOD = 25,000/12.88/50,000,000 = 0.000039 human-years/kg
EFU-M_FOOD = 15,000/30/50,000,000 = 0.00001 human-years/kg

TEFU_FOOD_total ≈ 2,850 EFU/year
Allocated = 300 × 0.134 = 40 EFU/year
SS_FOOD = 2,850 – 40 = +2,810 EFU/year (Level 2: BALANCED)

Result: EFU-FOOD BALANCED
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VII. Implementation Notes

1. **Branding**: EFU-FOOD labels only after documented EFU audit (104.3 compliance)
2. **Farm-to-Fork alignment**: TEFU_FOOD metric supports food waste reduction targets
3. **Data sources**: Tier 1 = water meters, steam meters, energy bills; Tier 2 = production logs